

YEAR 7 • UNIT 3 • EXTRA HOMEWORK

The Point Never Moves

Reading decimals • multiplying and dividing by 10 • rounding • mg, g, kg and t

Name: _____

Class: _____

Date given: _____

Date due: **Monday 28 September 2026****Why this sheet exists**

0.46, 4.6, 46 and **460** are the **same two digits**. Nothing about the 4 or the 6 is different. What tells you which number you are looking at is **where the point is**.

So when somebody says “multiply by 100”, the point **stays exactly where it is** and the **digits move** two columns to the left. That is why 7.2×10^3 is 7200 and not 7.2000 – and you can watch it happen in the tables on this sheet.

No calculator. Write in every box.

Part 1 • Where the point is

X1 One digit per column

Put each number into the table. One digit goes in each column. The first one is done for you.

hundreds	tens	ones	.	tenths	hundredths	thousandths
47.3						
	4	7	.	3		
6.85						
			.			
130.06						
			.			
0.904						
			.			

X2 The same digit, five different jobs

Every number below has a **6** in it, and the 6 is worth something different every time. Join each number to what its 6 is worth. The first one is done for you.

Word help – the columns after the point

tenths the **first** column after the point. 0.6 is six tenths

hundredths the **second** column. 0.06 is six hundredths

thousandths the **third** column. 0.006 is six thousandths

Number		What the 6 is worth
1. 6.2	C	A six tenths
2. 0.63		B six hundredths
3. 0.86		C six ones
4. 0.006		D six tens
5. 62.5		E six thousandths

X3 Saying it out loud**After the point, one digit at a time**

12.34 is “**twelve point three four**”. **Not** “**twelve point thirty-four**”.

The digits after the point are read **one by one**, so a zero has to be said: 4.05 is “**four point zero five**”.

Number	Say it
12.34	twelve point three four
5.9	
0.71	
8.06	
	three point one four
	twenty point zero five

X4 Which is bigger?

Write < or > in each box.

A longer number is not a bigger number

0.65 has more digits than 0.7, and it is **smaller**. Compare the **tenths first**: 6 tenths against 7 tenths. So $0.65 < 0.7$.

(a) $0.4 \square 0.35$

(c) $2.5 \square 2.45$

(e) $7.30 \square 7.3$

(b) $0.8 \square 0.81$

(d) $0.09 \square 0.1$

(f) $0.62 \square 0.6$

One pair in that list is **neither** bigger nor smaller. Which one, and why?

Part 2 · Moving the digits
X5 Watch it happen

Each row is the row above it multiplied by **10**. Every digit moves **one column to the left**. Fill in the rest of the table.

hundreds	tens	ones	.	tenths	hundredths	thousandths
		4	.	6		
	4	6	.			
			.			

(a) Write the three numbers. 4.6, _____, _____

(b) The 4 and the 6 are in all three rows. What moved, and what stayed still?

X6 Multiplying moves the digits left

One place for 10, two for 100, three for 1000.

(a) $3.2 \times 10 = \underline{\hspace{2cm}}$

(d) $3.2 \times 100 = \underline{\hspace{2cm}}$

(g) $3.2 \times 1000 = \underline{\hspace{2cm}}$

(b) $0.7 \times 10 = \underline{\hspace{2cm}}$

(e) $0.7 \times 100 = \underline{\hspace{2cm}}$

(h) $0.05 \times 100 = \underline{\hspace{2cm}}$

(c) $5.48 \times 10 = \underline{\hspace{2cm}}$

(f) $6.9 \times 100 = \underline{\hspace{2cm}}$

(i) $1.28 \times 1000 = \underline{\hspace{2cm}}$

X7 Dividing moves the digits right

The zeros you have to write yourself

$7 \div 100$. The 7 moves two columns right, past the ones and past the tenths, and lands in the **hundredths**. Nothing is left in the ones or the tenths, so you write **0** in both:
 $7 \div 100 = 0.07$.

- (a) $46 \div 10 =$ _____ (d) $46 \div 100 =$ _____ (g) $46 \div 1000 =$ _____
 (b) $8 \div 10 =$ _____ (e) $8 \div 100 =$ _____ (h) $9 \div 1000 =$ _____
 (c) $520 \div 10 =$ _____ (f) $520 \div 100 =$ _____ (i) $3.5 \div 100 =$ _____

X8 The little raised number**The power counts the zeros**

10^2 means $10 \times 10 = 100$. Two zeros. 10^3 means $10 \times 10 \times 10 = 1000$. Three zeros.
 Say it aloud: 10^2 is “**ten squared**”, 10^3 is “**ten cubed**”, 10^4 is “**ten to the power of four**”.

Power	Written out	Say it
10^2	100	ten squared
10^3		
10^5		
	10 000	

X9 Now with the power written in

Same job as X6 and X7. The power tells you **how many places**.

The folk rule is wrong

“To multiply by 10, add a zero” works for whole numbers and **nothing else**. 7.2×10^3 is **7200**. **Not 7.2000, which is just 7.2.**

- (a) $2.4 \times 10^2 =$ _____ (d) $830 \div 10^2 =$ _____
 (b) $0.06 \times 10^3 =$ _____ (e) $6 \div 10^3 =$ _____
 (c) $15 \times 10^4 =$ _____ (f) $2.5 \div 10^1 =$ _____

X10 Which way, and how far?

Look at what happened to the digits. Write **left** or **right**, and how many places. The first one is done for you.

From		To	Left or right?	How many places?	So you
4.8	→	480	left	2	$\times 10^2$
6.1	→	0.061			
90	→	9000			
7	→	0.7			

Part 3 · Stopping somewhere**X11 Which end is it nearer?**

The line below runs from 3 to 4, and it is marked in **tenths**.



(a) Write the number that **P** is marking. _____

(b) Write the number that **Q** is marking. _____

(c) Is **P** nearer to 3 or to 4? _____

(d) Is **Q** nearer to 3 or to 4? _____

X12 Round to the nearest whole number

Look at the next digit only

Look at the **tenths** digit. 0, 1, 2, 3, 4 – round **down**. 5, 6, 7, 8, 9 – round **up**.

12.3 → **12** 12.8 → **13** 12.5 → **13**

(a) 6.2 → _____

(c) 14.5 → _____

(e) 29.8 → _____

(b) 8.9 → _____

(d) 0.4 → _____

(f) 0.61 → _____

X13 Counting decimal places

Decimal places are the digits **after the point**. Start counting **at the point**, not at the front of the number.

Number	How many d.p.?	Number	How many d.p.?
5.28	2	130.4	
0.9		7.0625	
12.346		86	

X14 Round to 1 decimal place

Keep **one** digit after the point. Look at the **second** digit after the point to decide.

- (a) 4.62 → _____ (c) 0.348 → _____ (e) 9.44 → _____
 (b) 8.17 → _____ (d) 23.75 → _____ (f) 0.251 → _____

X15 The zero that has to stay

35 and 35.0 are not the same answer

34.98 to **1 decimal place**. The 8 rounds the 9 up, the 9 carries into the 4, and you get **35.0**.

Write **35.0**, not 35. The **.0** is what shows you rounded to **one** place — and it is the whole mark.

Round each one to **1 decimal place**. Every answer on this line ends in a zero.

- (a) 6.97 → _____ (c) 19.98 → _____
 (b) 0.96 → _____ (d) 2.04 → _____

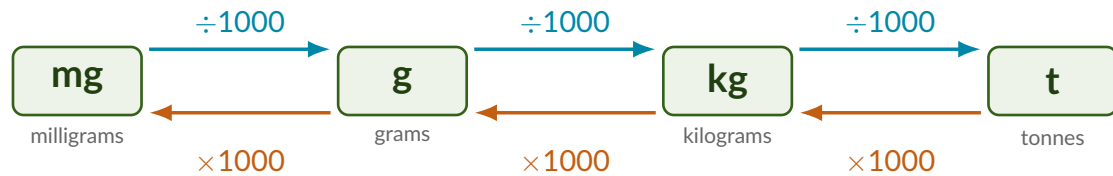
X16 Round to 2 decimal places

Keep **two** digits after the point. Look at the **third**.

- (a) 3.472 → _____ (c) 12.005 → _____
 (b) 0.1849 → _____ (d) 8.996 → _____

Part 4 • The mass ladder

There are four units of mass and they are **three identical steps** apart. Every step is $\times 1000$ one way and $\div 1000$ the other – which is the digits moving **three places**, exactly as in Part 2.



X17 One step

Use the ladder. Each of these is **one** step.

- | | |
|-----------------------|----------------------|
| (a) 5 kg = _____ g | (d) 750 g = _____ kg |
| (b) 2000 mg = _____ g | (e) 1.8 g = _____ mg |
| (c) 0.4 t = _____ kg | (f) 60 kg = _____ t |

X18 Two steps

Two steps is $\times 1000$ twice

kg to mg goes **kg** $\rightarrow$ **g** $\rightarrow$ **mg**: $\times 1000$, then $\times 1000$ again. That is $\times 10^6$ – the digits move **six** places.

$$4 \text{ kg} = 4000 \text{ g} = 4\,000\,000 \text{ mg}$$

- | | |
|---------------------|---------------------------|
| (a) 3 kg = _____ mg | (c) 500 000 mg = _____ kg |
| (b) 2 t = _____ g | (d) 0.25 t = _____ g |

X19 Which is heavier?

You cannot compare two numbers until they are in the **same unit**. Change one of them, then write $<$ or $>$.

- (a) 400 g or 0.5 kg ?

same unit: _____

☐

- (b) 2500 mg or 2 g ?

same unit: _____

☐

- (c) 0.9 t or 950 kg ?

same unit: _____

☐

X20 Mr Bowen's homework

Only **one** of these six is right. Find it, and correct the other five.

- (a) $4.6 \times 100 = 4.600$ _____ (d) 130.4 has 4 d.p. _____
(b) $7 \div 100 = 0.7$ _____ (e) $3 \text{ kg} = 3000 \text{ mg}$ _____
(c) 0.96 to 1 d.p. = 1.0 _____ (f) $0.65 > 0.7$ _____

X21 The whole sheet in one question

Mr Bowen's medicine bottle says **0.045 kg**.

- (a) Write that mass in **grams**. _____
(b) Write it in **milligrams**. _____
(c) Round your **gram** answer to **1 decimal place**. _____

Show the steps of the ladder here.

(d) The digits **4** and **5** are in all three of your answers, and nothing else about them is the same. Say in one or two sentences what changed.

Answer Key – Teacher Copy

X1 One digit per column

6.85: 6 in ones, 8 in tenths, 5 in hundredths.

130.06: 1 in hundreds, 3 in tens, 0 in ones, 0 in tenths, 6 in hundredths.

0.904: 0 in ones, 9 in tenths, 0 in hundredths, 4 in thousandths.

130.06 is the row that does the work, and the two zeros are why it is on the sheet. The 0 in the ones column and the 0 in the tenths column are both *holding a place open*. A student who leaves them out writes 13.6 and has changed the number by more than a hundred.

0.904 traps the other way: there is nothing in front of the point, and students write 904 or leave the ones column blank. The 0 in the ones is written, and the table has a column for it.

X2 The same digit, five different jobs

1 C · 2 A · 3 B · 4 E · 5 D

2 and 3 are the pair to mark together: 0.63 and 0.86 both have a 6 after the point, and it is worth ten times more in one than the other. Nothing about the digit says which — only the column does.

5 is the one students skip, because the 6 is in front of the point and they have decided the question is about decimals.

X3 Saying it out loud

5.9 five point nine · 0.71 nought point seven one · 8.06 eight point zero six

three point one four is 3.14 · twenty point zero five is 20.05

This is the listening question and it is worth doing out loud with the class. “Twelve point thirty-four” is what almost everyone says first, and it is wrong in a way that matters: “point thirty-four” would have to mean the tens column after the point, and there is no such column.

8.06 and 20.05 are the two that decide it. A student who writes 8.6 from “eight point zero six” has not heard the zero, and a student who says “eight point six” for 8.06 has not read it. Same error, two directions.

Accept “zero” or “nought” for 0. Do not accept “oh” in written work.

X4 Which is bigger?

(a) > (b) < (c) > (d) < (e) equal (f) >

The pair that is neither: (e), 7.30 and 7.3. A zero on the *end* of a decimal adds nothing — there are three tenths and no hundredths either way.

(a), (c) and (f) are all the same question: the longer number is the smaller one. Students compare digit counts instead of comparing tenths.

(d) is the hardest, because 0.09 has a 9 in it and 0.1 has a 1, and 9 is bigger than 1. It is in the *hundredths* column, so it is worth less. Send them to the table in X1.

(e) is the one that matters for Part 3. A trailing zero changes nothing about the size of a number — which is exactly why X15’s 35.0 is not about size. It is about saying how exact you were.

X5 Watch it happen

Row 2 is 46· (4 in tens, 6 in ones); row 3 is 460· (4 in hundreds, 6 in tens, 0 in ones).

(a) 4.6, 46, 460.

(b) The **digits** moved, one column to the left each time. The **point** stayed where it was.

(b) is the sheet. If a student can write that sentence, the rest of Part 2 is bookkeeping. If they cannot, no amount of practice at X6 will help, because they are still adding zeros.

Row 3 needs a 0 written in the ones column. It is the first time on the sheet that a zero has to be invented rather than copied, and it is the same zero as $7 \div 100 = 0.07$ in X7.

X6 Multiplying moves the digits left

(a) 32 (b) 7 (c) 54.8 (d) 320 (e) 70 (f) 690
(g) 3200 (h) 5 (i) 1280

(b) and (h) are the two that look wrong to a student: 0.7×10 is 7, with nothing after the point at all, and 0.05×100 is 5. Multiplying made the point disappear. It did not — the digits walked past it.

(a), (d) and (g) are deliberately the same digits three times. Point at the column.

X7 Dividing moves the digits right

(a) 4.6 (b) 0.8 (c) 52 (d) 0.46 (e) 0.08 (f) 5.2
(g) 0.046 (h) 0.009 (i) 0.035

(e), (g), (h) and (i) are the whole point of the question: a zero has to be *written* where nothing landed. Expect 0.8 for (e), 0.46 for (g) and 0.9 for (h) — in each case the student moved the right number of places and then declined to write the zeros that prove it.

(i) is the hardest item on the sheet: 3.5 already has a decimal point, and the digits still move two places right. $3.5 \rightarrow 0.35 \rightarrow 0.035$.

X8 The little raised number

$10^3 = 1000$, ten cubed $\cdot 10^5 = 100\,000$, ten to the power of five $\cdot 10^4 = 10\,000$, ten to the power of four

The third column is the question. “Squared” and “cubed” are not optional vocabulary — they are how the exam paper says 10^2 and 10^3 , and from 10^4 upwards English switches to “to the power of”. Make them say all four aloud.

The last row runs backwards, from the written-out number to the power. Students count the digits (5) instead of the zeros (4).

X9 Now with the power written in

(a) 240 (b) 60 (c) 150 000 (d) 8.3 (e) 0.006 (f) 0.25

(b) is the one to watch: 0.06×10^3 is 60, and it is a jump of three columns straight through the point. Expect 0.06000 from anyone still adding zeros, and 6 from anyone who moved two.

(f) is 10^1 , which the class has probably not seen written down. It is just 10. Say so; do not make a lesson of it.

X10 Which way, and how far?

$6.1 \rightarrow 0.061$: right, 2, $\div 10^2$

$90 \rightarrow 9000$: left, 2, $\times 10^2$

$7 \rightarrow 0.7$: right, 1, $\div 10^1$ (or $\div 10$)

This is X6 to X9 read backwards, and it is the question that shows whether a student is counting columns or guessing.

The first row is the one that goes wrong. 6.1 to 0.061 looks like “two zeros appeared”, so students write 3. Make them put both numbers in a place-value table and count where the 6 went: ones to hundredths is two columns.

X11 Which end is it nearer?

- (a) 3.4 (b) 3.7 (c) nearer 3 (d) nearer 4

The whole of X12 is on this picture, and it is worth saying so out loud: rounding is not a rule about the digit 5, it is a question about which end of the line you are nearer, and the digit is just the quick way to answer it.

If a student gets (c) or (d) wrong, they are reading the marks rather than the distances. Put a finger on P and walk it both ways.

X12 Round to the nearest whole number

- (a) 6 (b) 9 (c) 15 (d) 0 (e) 30 (f) 1

(d) is the item that gets left blank: 0.4 rounds to **0**, and students do not believe zero is an answer.

(e) is the carry — 29.8 goes to 30, not 29 and not 20.

(c) is the exactly-halfway convention. 14.5 rounds *up* because that is the rule, not because 15 is nearer; 14 and 15 are the same distance away. Say that once.

X13 Counting decimal places

5.28: 2 · 0.9: 1 · 12.346: 3 · 130.4: 1 · 7.0625: 4 · 86: **0**

130.4 is the trap and it is the reason the question exists: students count *all* the digits, or count from the front, and answer 4. Decimal places are counted from the point, rightwards, and there is one.

86 has **no** decimal places. Several will leave it blank.

X14 Round to 1 decimal place

- (a) 4.6 (b) 8.2 (c) 0.3 (d) 23.8 (e) 9.4 (f) 0.3

(c) and (f) both have a *third* decimal, and it is there to be ignored. Rounding to 1 d.p. looks at the second digit only: 0.348 is 0.3, not 0.35 and not 0.4. A student who rounds 0.348 to 0.35 and then to 0.4 has rounded twice, which is the error E3 on the main packet is built on.

(b) is the carry within the decimals: 8.17 goes to 8.2.

X15 The zero that has to stay

- (a) 7.0 (b) 1.0 (c) 20.0 (d) 2.0

All four answers end in .0 and the instruction line says so, which turns this from a trick into a drill. Anyone who writes 7, 1, 20 and 2 has understood the arithmetic perfectly and lost every mark, and that is worth saying plainly.

(b) is the biggest jump: 0.96 becomes 1.0, and the number in front of the point changes from 0 to 1.

This is the same fact as X4(e) turned around. There, the trailing zero added nothing to the *size*. Here it adds nothing to the *size* either — what it adds is the statement *I rounded this to one place*, which is what the question asked for.

X16 Round to 2 decimal places

- (a) 3.47 (b) 0.18 (c) 12.01 (d)
- 9.00**

(d) is the hardest item in Part 3: 8.996 carries through both decimals and into the ones, and the answer is 9.00 — a number that looks nothing like the question. Expect 8.99, 9 and 9.0; only 9.00 is right.

(c) is the other carry: the 5 rounds the second 0 up to 1.

X17 One step

- (a) 5000 g (b) 2 g (c) 400 kg (d) 0.75 kg (e) 1800 mg (f) 0.06 t

(d) and (f) go *up* the ladder into a number smaller than 1, which is the direction nobody practises. Expect 750 000 for (d) and 60 000 for (f) — the student used the ladder and went the wrong way along it.

Ask which way before asking how far. Going up towards tonnes, the number gets smaller; going down towards milligrams, it gets bigger. That is one sentence and it fixes most of this question.

X18 Two steps

- (a) 3 000 000 mg (b) 2 000 000 g (c) 0.5 kg (d) 250 000 g

A student who writes 3000 for (a) has taken one step and stopped, and that is the only error worth naming here. Make them say “kg, g, mg” out loud and count two arrows before they write anything.

(d) has a decimal going down the ladder: 0.25 t is 250 kg is 250 000 g. The point disappears on the first step, which students distrust — it is X6(h) again.

X19 Which is heavier?

- (a) 0.5 kg is 500 g, so $400 < 500$: the **0.5 kg** is heavier.
 (b) 2 g is 2000 mg, so $2500 > 2000$: the **2500 mg** is heavier.
 (c) 0.9 t is 900 kg, so $900 < 950$: the **950 kg** is heavier.

Either unit is fine as the common one — accept 400 g against 500 g or 0.4 kg against 0.5 kg. What is not fine is comparing 400 with 0.5 and announcing that 400 is bigger, which is what happens if the conversion is skipped, and which gets (a) and (c) wrong together.

(b) is the one where skipping the conversion accidentally gives the right answer. Do not let it pass: ask for the working.

X20 Mr Bowen’s homework

- (a) **wrong** — 460. The digits move two columns left; you do not write zeros on the end of a decimal.
 (b) **wrong** — 0.07. Two places right from 7, and both empty columns have to be written as zeros.
 (c) **right**.
 (d) **wrong** — 130.4 has **1** decimal place. Count from the point, not from the front of the number.
 (e) **wrong** — 3 000 000 mg. kg to mg is two steps, not one.
 (f) **wrong** — $0.65 < 0.7$. Six tenths against seven tenths.

One trap each from X6, X7, X15, X13, X18 and X4, in that order, which makes this a five-minute diagnosis of the whole sheet. Whichever one a student gets wrong, send them back to that question rather than explaining it again.

The instruction says one of the six is right. A student who corrects all six has not read it, and a student who corrects none has decided Mr Bowen is reliable.

(c) is the right one on purpose: it is the answer that *looks* wrong, because 1.0 looks like it should just be 1. That is X15, and it is the habit most likely to have slipped back.

X21 The whole sheet in one question

- (a) **45 g** (b) **45 000 mg** (c) **45.0 g**
 (d) Nothing changed about the digits. Only **where the point is** — and so which column each digit is standing in.
 (a) is three places left, and the leading zero goes: $0.045 \rightarrow 0.45 \rightarrow 4.5 \rightarrow 45$. Expect 0.045000 and 45 000.
 (c) is the question inside the question. 45 has no decimal places, so rounding it to 1 d.p. gives **45.0** — a trailing zero on a number that never had a decimal part at all. It is X15 with the carry taken away, and it is the last thing on the sheet on purpose.
 (d) has no single right wording. Accept anything that says the digits stayed and the point moved — or, better, that the *digits* moved and the *point* stayed, which is what the cover claims and what X5(b) asked for. Both sentences describe the same thing and a student who can argue for either has understood it.